Chapter 1 solutions

CH01-E1

Equating components gives

$$2c + d = 5, \qquad c + 2d = 4.$$

Twice the first equation minus the second gives

$$4c + 2d - (c + 2d) = 10 - 4,$$

Thus $3c = 6$ and $c = 2$. Substitution into the first equation gives $d = 1$.

The coefficient matrix

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

has determinant $2 \cdot 2 - 1 \cdot 1 = 3 \neq 0$, so the solution is unique.

CH01-E2

The inner product is

$$u^T v = 3 \cdot 4 + 4 \cdot 3 = 24.$$

Both norms are 5, so

$$\cos\theta = \frac{24}{25}.$$

Cauchy–Schwarz states

$$|u^T v| \leq \|u\|_2 \|v\|_2.$$

Here $24 < 25$. The strict inequality agrees with the fact that these vectors are not collinear.

A vector $z = (z_1, z_2)^T$ orthogonal to u must satisfy

$$3z_1 + 4z_2 = 0.$$

Taking $z_1 = 4$ gives $z_2 = -3$, so $z = (4, -3)^T$ is one nonzero answer. All solutions are

$$z = t(4, -3)^T, \qquad t \in \mathbb{R},$$

the line through the origin perpendicular to u. To restrict the set to nonzero vectors, require $t \neq 0$.

CH01-E3

Column view:

$$A(2, -1)^T = 2(1, 3)^T - (2, 4)^T = (0, 2)^T.$$

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Row view:

The first row gives $(1,2)\cdot(2,-1)=2-2=0$. The second gives $(3,4)\cdot(2,-1)=6-4=2$. Hence the output is again $(0,2)^T$.

The two decompositions compute the same vector.

CH01-E4

Since $a_2=2a_1$, the nonzero coefficient vector $(c_1,c_2)=(2,-1)$ satisfies

$$2a_1-a_2=0.$$

The vectors are dependent and span the line

$$\{t(1,2)^T : t\in\mathbb{R}\}.$$

For $(3,5)^T$ to belong to this span, its second component would have to be twice its first. Since $5\neq 2\cdot 3$, there is no solution.

For the target $(3,6)^T$, every solution satisfies

$$x_1+2x_2=3.$$

Set $x_2=t$. Then $x_1=3-2t$, and the complete solution set is

$$\left\{\begin{bmatrix}3\\0\end{bmatrix}+t\begin{bmatrix}-2\\1\end{bmatrix} : t\in\mathbb{R}\right\}.$$

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CH01-E5

The determinant is

$$\det A=2\cdot 2-1\cdot 1=3\neq 0.$$

Over the real field, a square matrix with nonzero determinant is invertible. The 2×2 formula gives

$$A^{-1}=\frac{1}{3}\begin{bmatrix}2&-1\\-1&2\end{bmatrix}.$$

Verification gives

$$\begin{bmatrix}2&1\\1&2\end{bmatrix}\left(\frac{1}{3}\begin{bmatrix}2&-1\\-1&2\end{bmatrix}\right)=\frac{1}{3}\begin{bmatrix}3&0\\0&3\end{bmatrix}=I_2.$$

Matrix multiplication does not commute in general. A matrix and its inverse nevertheless multiply to I_2 in both orders.

CH01-E6

A is 2×3, so $x\in\mathbb{R}^3$ and $Ax\in\mathbb{R}^2$. The column combination is

$$1(1,4)^T - (2,5)^T + 2(3,6)^T = (5,11)^T.$$

The row dot products are $1 - 2 + 6 = 5$ and $4 - 5 + 12 = 11$. Both group the same terms $a_{ij}x_j$.

CH01-E7

1. The coordinate vectors are

$$[p_1]_B = (1,2,0)^T, \quad [p_2]_B = (1,-1,0)^T, \quad [p_3]_B = (0,1,1)^T.$$

Thus

$$D = \begin{bmatrix} 1 & 1 & 0 \\ 2 & -1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

Expansion along the third row gives $\det D = (+1)\det \begin{bmatrix} 1 & 1 \\ 2 & -1 \end{bmatrix} = -3 \neq 0$, so the coordinate columns and the three polynomials are independent.

The coordinates of q are $(4,3,-1)^T$. Solve

$$c_1 + c_2 = 4, \qquad 2c_1 - c_2 + c_3 = 3, \qquad c_3 = -1.$$

The first equation gives $c_2 = 4 - c_1$. Substitution into the second gives $2c_1 - (4 - c_1) - 1 = 3$, hence $3c_1 = 8$, $c_1 = 8/3$, and $c_2 = 4/3$. Therefore

$$q = \frac{8}{3}p_1 + \frac{4}{3}p_2 - p_3.$$

Expanding verifies the constant term $8/3 + 4/3 = 4$, the coefficient of t, $16/3 - 4/3 - 1 = 3$, and the coefficient -1 of t^2.

2. The column-major coordinate columns are

$$(1,0,0,1)^T, \qquad (1,0,1,0)^T, \qquad (0,1,1,0)^T.$$

If $c_1A_1 + c_2A_2 + c_3A_3 = 0$, the entries give $c_1 + c_2 = 0$, $c_2 + c_3 = 0$, $c_3 = 0$, $c_1 = 0$, so the matrices are independent. However, $M_2(\mathbb{R})$ has dimension 4. Three vectors cannot span this four-dimensional space, so the independent set is not a basis.

CH01-E8

1. For $Av = \lambda v$, $v \neq 0$, induction gives $A^k v = \lambda^k v$, hence $r(A)v = r(\lambda)v$ for every polynomial. Conversely, let $\mu \in \operatorname{spec}(r(A))$. For nonconstant r, factor $r(t) - \mu = c\prod_j (t - \zeta_j)$ over $\mathbb{C}$. From

$$0 = \det(r(A) - \mu I) = c^n \prod_j \det(A - \zeta_j I)$$

at least one ζ_j is an eigenvalue of A, with $r(\zeta_j) = \mu$. A constant polynomial is handled by $r(A) = cI$. Thus $\operatorname{spec}(r(A)) = r(\operatorname{spec}(A))$, without diagonalization.

Write $J = 3I + N$, where

$$N = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \qquad N^2 = 0.$$

For every positive integer k, $J^k = 3^k I + k3^{k-1}N$. The polynomial $r(t) = t^3 - 2t + 4$ therefore gives

$$r(J) = r(3)I + r'(3)N.$$

This follows directly from the binomial expansion. If $Ju = 3u$, each term satisfies $J^k u = 3^k u$, so $r(J)u = r(3)u = 25u$.

2.

$$r'(t) = 3t^2 - 2, \qquad r'(3) = 25,$$

Also $r(3) = 27 - 6 + 4 = 25$. Hence

$$r(J) = 25I + 25N = 25 \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}.$$

3. No. The counterexample $\begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$ has only eigenvalue 2, but its eigenvectors lie in the line spanned by e_1, and the matrix is not $2I$. The eigenvalue determines the diagonal of a Jordan block. Without a full independent eigenbasis, the generalized-eigenvector structure can retain nonzero off-diagonal entries.

References and attribution

Lin Cing, handwritten linear algebra notes. Gilbert Strang, Introduction to Linear Algebra, 5th ed., Chapter 1.
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https://github.com/xinhi6716/computer-science-self-study